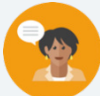


7.3 Factoring a Trinomial With a Leading Coefficient of 1

Lesson Introduction				
 Benchmark	MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system.		 Learning Target	I can factor a trinomial with a leading coefficient of 1.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.		MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real- or complex number system.	
 Essential Questions	- What methods can you use to factor a trinomial with a leading coefficient of one? - What processes can be used to factor a trinomial?			
 Lesson Narrative	In this lesson, students continue their learning of factoring trinomials with a leading coefficient of one. Students will begin by collaborating on different methods for factoring trinomials, including using the area model. They will have the opportunity to learn how to factor trinomials with a leading coefficient of one using different methods and end the lesson by completing the factoring with the method of their choice.			
 Component Summary	7.3.1 Warm-Up (~5 min.)	Factor a trinomial (<i>SE p. 370</i>)	7.3.2 Collaboration (15-20 min.)	Explore different methods of factoring, including the area model and factoring by grouping (<i>SE p. 370</i>)
	7.3.3 Guided Instruction (10-15 min.)	Formal instruction on factoring trinomials where $a = 1$ (<i>SE p. 373</i>)	7.3.4 Your Turn! (10 min.)	Independent practice factoring trinomials using preferred method (<i>SE p. 374</i>)
	7.3.5 Wrap-Up (~5 min.)	Lesson summary writing activity (<i>SE p. 374</i>)		
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> - Area model - Quadratic expression (equation) <u>Additional Terms:</u> N/A		N/A	

 Teacher Tips	Common Misconceptions • Students may believe only one method of factoring can be used in specific situations. • Students may attempt to factor without considering the a term in quadratic expressions.	Things to Consider • This lesson is a continuation of student learning in the prior lesson. Students are going to explore the methods of the area model and factoring by grouping. Pay close attention to how students perform in each method to help them determine which method best suits them.
	Literacy and Language Routines Compare and Connect 7.3.3 Guided Instruction provides students with formal instruction on factoring trinomials where $a = 1$. When teaching students the different methods, consider leveraging this routine to share your thinking while working through the problems in the component. This will support students in making connections between the various representations as they develop their understanding of factoring.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2:1 Multiple Representations 7.3.2 Collaboration introduces students to different methods for factoring trinomials with a leading coefficient of one. Students will explore multiple methods such as grouping and area model. In this component, students progress from modeling problems with objects and drawings to using abstract representations.
	 Differentiate Instruction	
Lesson Notes:		

7.3.1 Warm-Up: Bell Work

Component Overview: Students will complete a Bell Work activity where they factor a trinomial with a leading coefficient of one.



7.3.1 Warm-Up: Bell Work

1. Complete the table for $x^2 + 3x - 18$.

	x	-3
x	x^2	☐ $1x$ ☐ $-1x$ ☐ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $9x$ ☐ $-9x$ ☐ $3x$ ☒ $-3x$ ☐ $18x$ ☐ $-18x$
$+6$	☐ $1x$ ☐ $-1x$ ☒ $6x$ ☐ $-6x$ ☐ $2x$ ☐ $-2x$ ☐ $9x$ ☐ $-9x$ ☐ $3x$ ☐ $-3x$ ☐ $18x$ ☐ $-18x$	-18



Think-Pair-Share

This task is identical to what students did in the prior lesson. Allow students to have the opportunity to try this one on their own to see what they recall. Then, have students work with a partner to share their answers and come to a consensus. Once partners have had an opportunity to discuss, facilitate a whole-class review of the correct answer if necessary and if time allows.

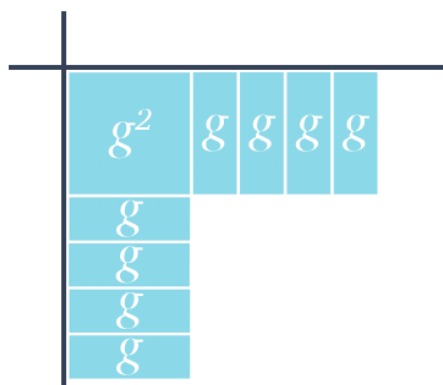
7.3.2 Collaboration: Different Methods of Factoring

Component Overview: Students will work with a partner or small group to explore different methods of factoring, including area models and grouping.



7.3.2 Collaboration: Different Methods of Factoring

1. Garcelle is trying to factor the trinomial $g^2 + 8g + 12$ using algebra tiles.



Discuss with your partner the mistake that Garcelle made. Write a note to Garcelle to explain her mistake and how she can fix it.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary. Hi Garcelle! Good job recognizing that $4g + 4g$ is $8g$. However, the current setup would require 16 positive one tiles, which won't match the trinomial. Can you think of factors of 12 that add up to be 8? That should help you get back on track!



MA.K12.MTR.3.1: Mathematical Fluency

Collaboration intentionally highlights three differing methods that produce the same result. Continue highlighting the role of flexibility in fluency, that different representations can provide more pathways that end with the same answer. In the 7.3.3 Guided Instruction, the teacher should continue using visual models if the students are struggling.



Question 1 is a recall back to what students did in the prior lesson. Use the data from that lesson to determine which student pairs you may need to visit first to ensure they are on the right track in this component.



How many unit tiles will be needed to complete the algebra-tile model Garcelle created? How could you "fix" the g -tiles so that the result will be only 12 unit tiles?

7.3.2 Collaboration: Different Methods of Factoring (continued)

2. Complete the table for $g^2 + 8g + 12$.

	g	$+2$
g	g^2	☐ $1g$ ☐ $-1g$ ☐ $4g$ ☐ $-4g$ ☐ $2g$ ☐ $-2g$ ☐ $6g$ ☐ $-6g$ ☐ $3g$ ☐ $-3g$ ☐ $12g$ ☐ $-12g$
$+6$	☐ $1g$ ☐ $-1g$ ☐ $4g$ ☐ $-4g$ ☐ $2g$ ☐ $-2g$ ☐ $6g$ ☐ $-6g$ ☐ $3g$ ☐ $-3g$ ☐ $12g$ ☐ $-12g$	12

3. What are the two binomial factors for $g^2 + 8g + 12$?

$$g^2 + 8g + 12 = (g + 2)(g + 6)$$

Felix showed Garcelle how to use the **area model** to factor $g^2 + 8g + 12$.



The **area model** is a rectangular diagram that utilizes the decomposition of side lengths by place value to multiply numbers using the distributive property.

4. Fill in the missing information for the **area model** for $g^2 + 8g + 12$.

	$1g$	6
$1g$	g^2	$6g$
2	$2g$	12



Monitor for students who are stuck or struggling in small groups and use inquiry to prompt discussion.

- How can you determine if the signs should be positive or negative?
- What is the relationship between the 12 and the number of g-tiles?
- Can the factors listed in the table be reversed to result in the same product?



Consider creating an anchor chart that represents a trinomial being factored using different methods but arriving at the same answer. Use different colors or other ways to distinguish one method from another easily.

The methods included could be using algebra tiles, the area model, or factoring by grouping.

7.3.2 Collaboration: Different Methods of Factoring (continued)

5. Garcelle asked Felix how to determine the monomial that belongs in the orange boxes when creating the area model for $k^2 - 3k - 28$. Felix said that he rewrites $k^2 - 3k - 28$ as $k^2 - 7k + 4k - 28$ because $-7 \cdot 4 = -28$ and $-7 + 4 = -3$. He knows that one of the orange boxes should result in the product $-7k$ and the other should result in the product $4k$.

- a. Complete the area model for $k^2 - 3k - 28$.

	<u>k</u>	<u>-7</u>
<u>k</u>	<u>k^2</u>	<u>$-7k$</u>
<u>4</u>	<u>$4k$</u>	<u>-28</u>

- b. What are the two binomial factors for $k^2 - 3k - 28$?

$$k^2 - 3k - 28 = (k + 4)(k - 7)$$

- c. Lori uses a different method of factoring. Like Felix, she rewrote $k^2 - 3k - 28$ as $k^2 - 7k + 4k - 28$, but instead paired the terms as shown.

$$(k^2 - 7k) + (4k - 28)$$

Work with your partner to finish factoring using Lori's method, known as **factoring by grouping**, using the following steps.

$(k^2 - 7k) + (4k - 28)$	
Factor out the greatest common factor from each pair of terms.	$k(k - 7) + 4(k - 7)$
The binomial, $k - 7$, is common so the sum can be rewritten.	$(k - 7)(k + 4)$



Consider allowing students to do a whole-class share out to discuss their findings and processes after completing all parts of question 5. Have them compare what they did with the area model in part a with their method of grouping in part c.

Consider creating a characteristics T-chart to have students share out features of both methods during their discussion.



If Lori grouped the expression as $(k^2 + 4k) + (-7k - 28)$, show if the factors will be the same.

7.3.2 Collaboration: Different Methods of Factoring (continued)

There are many different methods for factoring trinomials. Relating factoring to algebra tiles or the area model is useful for understanding why the factors of a trinomial of the form $ax^2 + bx + c$ are two binomials.

Using algebra tiles or the area model is also helpful in understanding the relationship the factors of the c -term have with the b -term.

6. Discuss with your partner the relationship the factors of the c -term have with the b -term based on the expressions explored in the previous problems.

Answers may vary. The b term is the sum of the factors of the c term.



*MA.K12.MTR.3.1: Mathematical Fluency
Different methods for factoring can be helpful for different students. Showing all methods helps to solidify understanding as well as provide insight into which methods make more or less sense to each of your students. Consider ways to leverage their different understandings to provide differentiated support through the remainder of this lesson and unit.*



Consider adding a debate activity here. Have students split into two groups. In one group, they are for using the area model, and the other group is for using the grouping method. Each team must provide reasoning and rationale on why their method is the best method to use when factoring trinomials.

7.3.3 Guided Instruction: Factoring Trinomials Where $a = 1$

Component Overview: Students will receive formal instruction on factoring trinomials where $a = 1$ using different methods.



7.3.3 Guided Instruction: Factoring Trinomials where $a = 1$



A **quadratic expression**, or **quadratic equation**, is a polynomial expression or equation that contains a term of degree 2, but no term of higher degree.

Factoring quadratic expressions in the form of trinomials such as $ax^2 + bx + c$, where $a = 1$, results in the factors $(x + g)(x + h)$ where, $g \cdot h = c$ and $g + h = b$. Later in this unit, factoring trinomials where a is a value other than 1 will be explored.

1. Consider the trinomials $y^2 + 14y + 24$, $y^2 - 14y + 24$, $y^2 + 10y - 24$, and $y^2 - 10y - 24$.

a. Factor each trinomial.

$y^2 + 14y + 24$	$y^2 - 14y + 24$	$y^2 + 10y - 24$	$y^2 - 10y - 24$
$(y + 2)(y + 12)$	$(y - 12)(y - 2)$	$(y - 2)(y + 12)$	$(y - 12)(y + 2)$

- b. Describe any similarities between the four trinomials.

All four trinomials have a constant value of 24 or -24.

- c. Describe any similarities between the factored forms of the four trinomials.

All four binomial factors have the numbers 12 and 2 as factors.



Teachers are encouraged to continue using different methods of factoring throughout this component while continuing to avoid “tricks” or shortcuts. Some students with a strong knowledge of factors will be able to determine the correct pair of factors without scaffolding. Other students will need scaffolding. Avoid insisting on only one method for scaffolding, as this can leave students without a method that supports their thinking.



Consider splitting the class based on their preferences for factoring from the prior component. Then they can work in partners to attempt part a before finding a partner from a different group and comparing and contrasting both their process and their product.

7.3.3 Guided Instruction: Factoring Trinomials Where $a = 1$ (continued)

2. Consider the trinomial $x^2 - 2x - 8$.

a. Factor $x^2 - 2x - 8$ using the area model.

	x	-4
x	x^2	$-4x$
-2	$-2x$	-8

b. Factor $x^2 - 2x - 8$ using factoring by grouping.

$$\begin{aligned}
 &x^2 - 4x + 2x - 8 \\
 &(x^2 - 4x) + (2x - 8) \\
 &x(x^2 - 4x) + 2(2x - 8) \\
 &x(x - 4) + 2(x - 4) \\
 &(x - 4)(x + 2)
 \end{aligned}$$

c. Discuss with your partner how to check that the two binomials of the factored form are correct.

The distributive property can be applied to the factored binomials to check if the product is the same as the original trinomial.



Consider creating a space on the whiteboard, chart paper, or other central location to show how this trinomial was factored using more than one method. Allow different students to come up to the board to do the first step of one method and have them “tag” another student to do the next step. Repeat this again for the other method until it is also factored. Then ask students to compare similarities and differences between the two methods.



Tricks and shortcuts lead to misconceptions. Avoid having students memorize phrases like “factors of c whose sum is b ” and instead use area model and grouping to represent their thinking. While students can use a strategy like finding factors of $(a)c$ whose sum is b , this approach should not be a driving instructional focus.

7.3.4 Your Turn!

Component Overview: Students will practice factoring quadratic expressions using the method of their choice. For this activity, allow the students to complete independently to practice what they've learned.



7.3.4 Your Turn!

1. Factor each quadratic expression using the method of your choice.

a. $d^2 + 11d + 24$
 $(d + 3)(d + 8)$

b. $a^2 - 9a + 14$
 $(a - 7)(a - 2)$

c. $x^2 + 4x - 96$
 $(x - 8)(x + 12)$



Consider having students find another student for each problem that factored it using a different method. With this partner, have them compare their answers and come to a consensus if their answers do not match. Once they've gotten their answers, have students initial that they've been used for that problem. Repeat this process for each part of the question to allow students the opportunity to work with different students throughout the review of this component.

7.3.5 Wrap-Up: Cool Down

Component Overview: Students will write a note to someone who missed class that summarizes the lesson. This activity is designed to be completed independently to allow students the opportunity to respond to an open-ended question about what they learned.



7.3.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class summarizing today's lesson. Include something you learned and a mistake you learned from.

Answers may vary.



Make space for creativity and choice in student responses to increase engagement and agency.

- *Students can use social media to record a short video describing their findings.*
- *Students can tutor a friend or family member and submit a recording of their session or evidence of the work of their friend or family member.*